

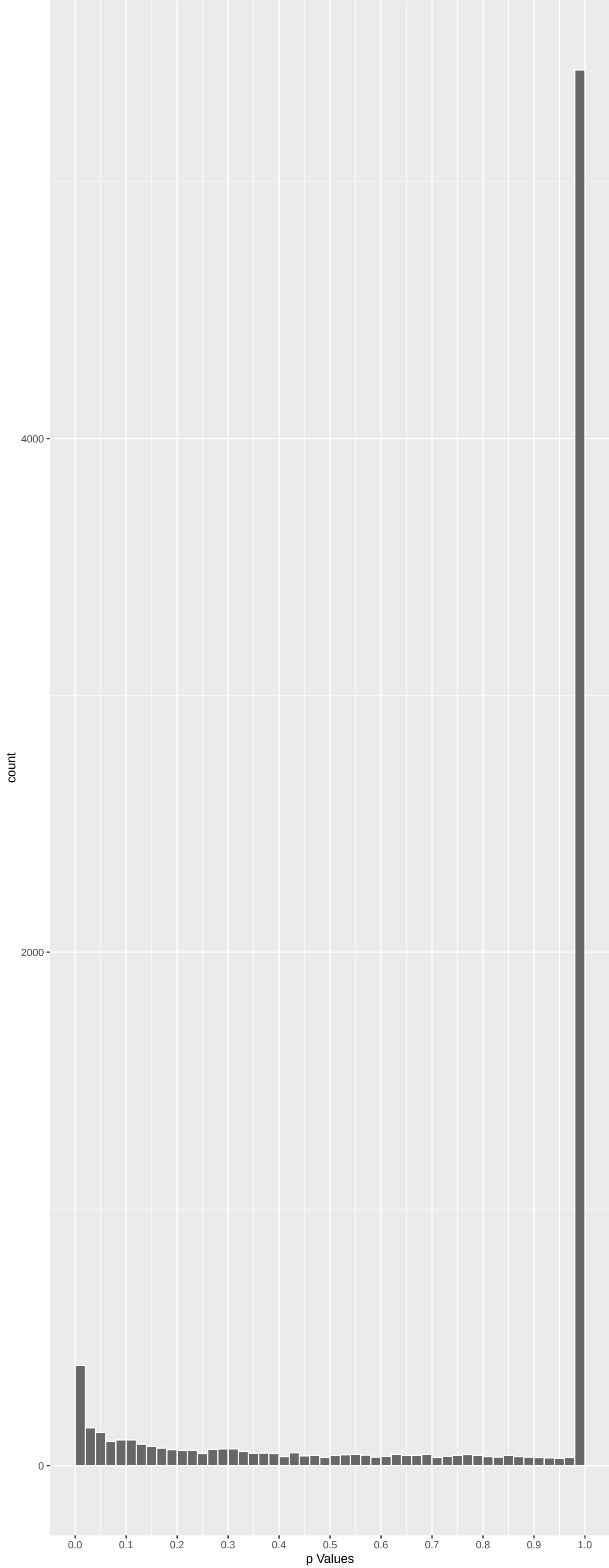
Top genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
CLDN16	5.986690	1.284926e-08	1.941e-04	1.000e+00
BPIFB1	5.850289	2.944313e-08	2.224e-04	1.000e+00
CLPS	5.024734	3.024789e-06	7.615e-03	1.000e+00
SLC23A1	5.038034	2.822028e-06	7.615e-03	1.000e+00
CPB1	5.059832	2.517755e-06	7.615e-03	1.000e+00
FAR2	5.065493	2.444072e-06	7.615e-03	1.000e+00
ST6GAL2	4.971277	3.990809e-06	8.612e-03	1.000e+00
EHHADH	4.786481	1.018386e-05	1.725e-02	1.000e+00
HMCES	4.784597	1.027985e-05	1.725e-02	1.000e+00
ALB	4.755973	1.184959e-05	1.790e-02	1.000e+00
GCDH	4.656781	1.927147e-05	1.927e-02	1.000e+00
RBSN	4.645199	2.038493e-05	1.927e-02	1.000e+00
SLC1A1	4.643276	2.057564e-05	1.927e-02	1.000e+00
EPN3	4.632435	2.168343e-05	1.927e-02	1.000e+00
CPAT2	4.698080	1.575711e-05	1.927e-02	1.000e+00
FGD5	4.646397	2.026692e-05	1.927e-02	1.000e+00
SPTA1	4.699564	1.564304e-05	1.927e-02	1.000e+00
PNLIP	4.612707	2.384754e-05	2.001e-02	1.000e+00
IL1R2	4.566210	2.979726e-05	2.369e-02	1.000e+00
COQ9	4.501934	4.040274e-05	3.051e-02	1.000e+00
SLCA4A1	4.469556	4.702936e-05	3.089e-02	1.000e+00
PLPP1	-4.475251	4.579297e-05	3.089e-02	1.000e+00
CHAC2	4.470329	4.685972e-05	3.089e-02	1.000e+00
HIRA	4.408623	6.241801e-05	3.735e-02	1.000e+00
TM4SF4	4.395087	6.643711e-05	3.735e-02	1.000e+00

Top genes by Q-Value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
M6PR	-0.22723602	1.000000000	1.000e+00	1.000e+00
FKBP4	1.48905881	0.818831294	1.000e+00	1.000e+00
CYP26B1	-1.22228561	1.000000000	1.000e+00	1.000e+00
NDUFAF7	-0.94274266	1.000000000	1.000e+00	1.000e+00
FUCA2	-0.01167970	1.000000000	1.000e+00	1.000e+00
DBND1	0.05102369	1.000000000	1.000e+00	1.000e+00
HS3T1	1.16817078	1.000000000	1.000e+00	1.000e+00
SEMA3F	1.06388121	1.000000000	1.000e+00	1.000e+00
CFTR	-0.98320336	1.000000000	1.000e+00	1.000e+00
CYP51A1	1.87343602	0.366048204	1.000e+00	1.000e+00
USP28	-0.43824354	1.000000000	1.000e+00	1.000e+00
SLC7A2	-0.15814565	1.000000000	1.000e+00	1.000e+00
HSPB6	-0.72199313	1.000000000	1.000e+00	1.000e+00
PK4	3.03067446	0.014640490	5.122e-01	1.000e+00
USH1C	-0.26189262	1.000000000	1.000e+00	1.000e+00
BAIAP2L1	2.35377380	0.1111503495	1.000e+00	1.000e+00
TMEM132A	0.51409698	1.000000000	1.000e+00	1.000e+00
DVL2	-3.14628613	0.009921489	4.295e-01	1.000e+00
RPAP3	-0.83778218	1.000000000	1.000e+00	1.000e+00
HCKA11	-1.70429835	0.529852179	1.000e+00	1.000e+00
TRAPPC6A	-0.11779667	1.000000000	1.000e+00	1.000e+00
SPATA20	-0.68470369	1.000000000	1.000e+00	1.000e+00
CD79B	0.43675758	1.000000000	1.000e+00	1.000e+00
RHBD1	2.60268517	0.055498111	9.608e-01	1.000e+00
RPUSD1	0.83047657	1.000000000	1.000e+00	1.000e+00

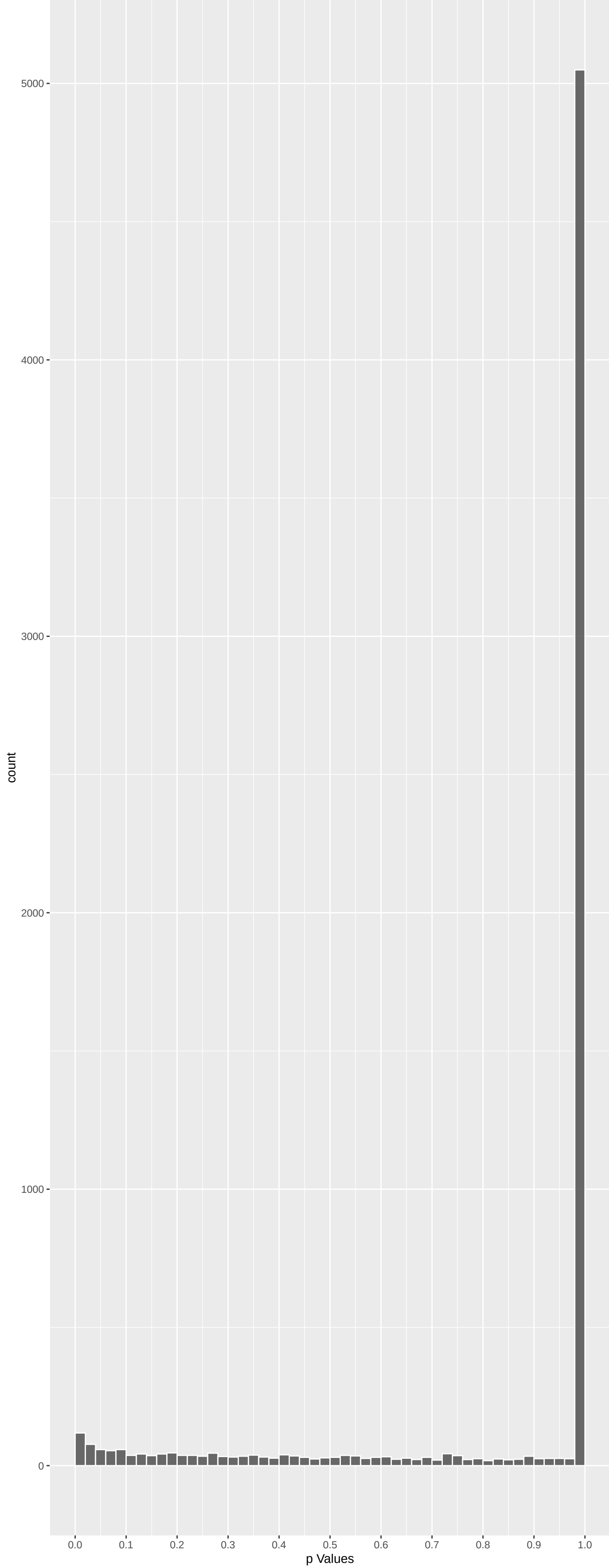
Positive Rho Non-permulated



Top Positive genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
CLDN16	5.986690	1.284926e-08	1.941e-04	1.000e+00
BPIFB1	5.850289	2.944313e-08	2.224e-04	1.000e+00
CLPS	5.024734	3.024789e-06	7.615e-03	1.000e+00
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CPB1	5.059832	2.517755e-06	7.615e-03	1.000e+00
FAR2	5.065493	2.444072e-06	7.615e-03	1.000e+00
ST6GAL2	4.971277	3.990809e-06	8.612e-03	1.000e+00
EHHADH	4.786481	1.018386e-05	1.725e-02	1.000e+00
HMCES	4.784597	1.027985e-05	1.725e-02	1.000e+00
ALB	4.755973	1.184959e-05	1.790e-02	1.000e+00

Negative Rho Non-permulated



Top Negative genes by P-value non-permulated

Gene	Rho	P	p.adj	qValueNoperm
PLPP1	-4.475251	4.579297e-05	3.089e-02	1.000e+00
BIRC3	-4.279569	1.123531e-04	4.714e-02	1.000e+00
MORN4	-4.255687	1.250447e-04	5.105e-02	1.000e+00
SNRPD3	-4.129321	2.183020e-04	7.016e-02	1.000e+00
RPL34	-4.062023	2.918952e-04	8.183e-02	1.000e+00
H2BC21	-4.032462	3.311739e-04	8.701e-02	1.000e+00
TRPC4AP	-4.007572	3.680770e-04	8.799e-02	1.000e+00
CFAP20	-3.970383	4.305429e-04	9.105e-02	1.000e+00
HSPB3	-3.970372	4.305629e-04	9.105e-02	1.000e+00
ICMT	-3.946960	4.748980e-04	9.661e-02	1.000e+00

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Inorganic Cation Transmembrane Transport	0.08675135	235	5.324e-06	1.614e-02	ANO6.95 SLC6A12.119 SLC36A2.138 HPN160 SLC6A15.194 ATP6V0A4.196
Monocarboxylic Cation Transmembrane Transport	0.08536139	234	7.845e-06	1.614e-02	ANO6.95 SLC6A12.119 SLC36A2.138 HPN160 SLC6A15.194 ATP6V0A4.196
Monocarboxylic Anion Transport (GO:001571)	0.18171165	50	9.015e-06	1.614e-02	SLC5A8.64 SLC5A12.67 SLC6A12.119 ABCD3.167 FABP1.179 KDM5A.401
Organic Hydroxy Compound Transport (GO:0)	0.20888495	36	1.468e-05	1.971e-02	SLC22A1.62 SLC5A8.64 ABCD3.167 GHR1.252 SLC5A12.67 SLC5A1.698
Carboxylic Acid Transport (GO:0046942)	0.16184370	47	1.259e-04	1.353e-01	SLC5A12.67 SLC6A12.119 SLC36A2.138 SLC6A15.194 SLC6A18.429 SLC6A6.478
Intracellular Monocarboxic Ion Homeostasis	0.2354165	19	4.269e-04	3.822e-01	CLDN16.103 CNMNA.204 SLC4A1.312 LETM1.510 SLC39A10.812 RHCG.813
Organic Anion Transport (GO:0015711)	0.11847502	72	2.513e-04	4.000e-01	SLC05A1.111 SLC52A3.143 SLC4A9.176 SLC4A1.312 SLC4A3.390 MGST1.402
Sodium Ion Transmembrane Transport (GO:0)	0.11890770	67	7.794e-04	5.232e-01	SLC6A12.119 SLC6A15.194 SLC9A4.319 SLC1A12.356 SCNN1B.358 SCNN1B.406
Organic Cation Transport (GO:0015695)	0.17407857	30	9.747e-04	5.817e-01	SLC22A1.62 SLC6A12.119 SLC18B1.126 SLC22A.675 SLC5A7.766 SLC4A45.769
Bile Acid And Bile Salt Transport (GO:00)	0.23220741	16	1.305e-03	5.843e-01	ABCD3.167 SLC51B.455 SLC51A.698 SLC10A2.869 SLC10A3.891 SLC27A5.1144
Nitrogen Compound Transport (GO:0071705)	0.08104306	133	1.301e-03	5.843e-01	SLC22A1.62 SLC5A8.64 SLC6A12.119 SLC36A2.138 SLC52A3.143 SLC6A15.194
Sulfur Compound Transport (GO:0072348)	0.20871490	20	1.238e-03	5.843e-01	SLC22A1.62 SLC26A4.296 SLC26A8.768 SLC13A1.798 SLC35B3.826 SLC25A10.848
L-phenylalanine Catabolic Process (GO:00)	0.42004141	4	3.621e-03	6.145e-01	TAT.546 GSTZ1.673 HGD.1024 IL4I1.1697 NA NA
L-phenylalanine Metabolic Process (GO:00)	0.42004141	4	3.621e-03	6.145e-01	TAT.546 GSTZ1.673 HGD.1024 IL4I1.1697 NA NA
Acid Secretion (GO:0046717)	0.31814180	7	3.561e-03	6.145e-01	DRD3.243 GHR1.252 SLC51B.455 SLC51A.698 SLC26A6.3075 SLC9A3R1.3295
Alpha-Amino Acid Biosynthetic Process (G	0.27832458	10	2.310e-03	6.145e-01	BCAT2.28 SDS1.134 CAD.730 SDS.847 CPS1.1137 BCAT1.2925
Canonical Wnt Signaling Pathway (GO:0060	-0.15408568	33	2.206e-03	6.145e-01	PTEN.5 WNT.4.25 DVL2.140 FRAT1.358 FZD6.407 BCL9L.411
Erythrose 4-Phosphate/Phosphoenolpyruvat	0.42004141	4	3.621e-03	6.145e-01	TAT.546 GSTZ1.673 HGD.1024 IL4I1.1697 NA NA
Ethanol Metabolic Process (GO:0006067)	0.34505051	6	3.425e-03	6.145e-01	ACSS2.412 ADH4.1182 SULT1A2.1487 SULT1E1.2135 SULT1A3.2950 SULT1B1.3463
Gamma-Aminobutyric Acid Transport (GO:00	0.38778691	5	2.676e-03	6.145e-01	SLC6A12.119 SLC6A6.478 SLC6A11.646 SLC6A13.2492 SLC9A3R1.3295 NA
Innate Immune Response In Mucosa (GO:000	0.42198546	4	3.468e-03	6.145e-01	BP1FB1.2 NOS2.197 PLA2G1B.225 IL4.3480 NA NA
Inorganic Cation Import Across Plasma Me	0.09640985	85	2.172e-03	6.145e-01	SLC12A9.271 CLU.299 SLC12A6.305 SLC9A4.319 SLC12A3.356 SCNN1G.358
Monocarboxic Cation Homeostasis (GO:005058	0.11232214	61	2.450e-03	6.145e-01	CLDN16.103 CNMNA.204 SLC12A9.271 SLC12A6.305 SLC12A3.356 SCNN1G.358
Mucosal Immune Response (GO:0002385)	0.28314470	10	1.935e-03	6.145e-01	BP1FB1.2 NOS2.197 PLA2G1B.225 FFA2.426 FNLR1.1338 RAB17.2714
Negative Regulation Of Gene Expression V	0.37637612	5	3.562e-03	6.145e-01	ZNF445.338 UHRF1.1311 USP7.1572 HELLS.1669 UHRF2.2788 NA
Negative Regulation Of miRNA Metabolic P	0.21677282	15	3.661e-03	6.145e-01	POU5F1.270 PDGFB.603 ESRI.1493 HF1A1.1560 NFATC4.1662 NFATC3.1690
Negative Regulation Of miRNA Transcripti	0.24167076	14	1.748e-03	6.145e-01	POU5F1.270 PDGFB.603 ESRI.1493 HF1A1.1560 NFATC4.1662 NFATC3.1690
Negative Regulation Of Single Stranded V	0.34850601	6	3.111e-03	6.145e-01	TASOR.5 MORC2.30 INPP5K.534 SETDB1.1016 MPHOSPH8.1841 HMGA2.8031.5
Positive Regulation Of DNA Methylation-D	0.31902278	8	1.782e-03	6.145e-01	TASOR.5 MORC2.30 SETDB1.1016 TRIM28.1724 MPHOSPH8.1841 PPHLN1.2236
Positive Regulation Of Heterochromatin F	0.26843169	10	3.249e-03	6.145e-01	TASOR.5 MORC2.30 SETDB1.1016 TRIM28.1724 MPHOSPH8.1841 PPHLN1.2236
Regulation Of Cell Adhesion (GO:0003155)	-0.08265646	105	3.441e-03	6.145e-01	TGM2.44 P4HB.70 RAC2.161 CXCR3.394 FUT2.400 FES.402
Sodium Ion Transport (GO:0008614)	0.10041948	74	2.872e-03	6.145e-01	SLC5A8.64 SLC5A12.67 SLC6A12.119 SLC6A15.194 SLC12A3.356 SCNN1G.358
One-Carbon Compound Transport (GO:001975	0.15367119	29	4.205e-03	6.643e-01	AQP11.35 SLC4A9.176 SLC4A1.312 SLC4A3.390 RHAG.855 CA2.947
Sulfate Transport (GO:0008272)	0.26202506	10	4.121e-03	6.643e-01	SLC26A4.296 SLC26A8.768 SLC13A1.798 SLC25A10.848 SLC26A2.2397 SLC13A4.2561
Chromosome Condensation (GO:00030261)	0.18648092	19	4.909e-03	7.202e-01	H1-6.139 AKAP81.527 PLK1.811 AIFM1.853 ERN2.564 NCAPH.1141
Fat-Soluble Vitamin Metabolic Process	0.20373483	16	4.793e-03	7.202e-01	CYP11A1.4 LRAF1.112 NQO1.374 KDM5A.401 GGX3.505 CYP11A1.695
Intracellular pH Elevation (GO:0051454)	0.30573074	7	5.095e-03	7.202e-01	NOX1.213 TMEM175.657 CLN3.833 SLC26A3.1580 OCA2.2165 SLC26A6.3075
Replication Fork Processing (GO:0031297)	-0.13348788	37	4.995e-03	7.202e-01	BRCA1.247 RAD51.284 MUS81.369 PARP1.4699.5 ATRX.4699.5 WRN.4699.5
Aromatic Amino Acid Family Catabolic Pro	0.26303706	9	6.292e-03	8.242e-01	TAT.546 GSTZ1.673 HPD.760 HGD.1024 IL4I1.1697 KMO.2841
Sodium Ion Homeostasis (GO:0050578)	0.19151404	17	6.278e-03	8.242e-01	SLC12A3.356 SCNN1G.358 SCNN1B.406 ATP1B3.477 SCNN1A.653 ATP4B.1118

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	0.12279528	219	4.404e-10	2.852e-06	SLC1A1.12 SLC22A1.62 SLC5A8.64 SLC5A12.67 SLC6A12.119 SLC36A2.138
REACTOME_TRANSPORT_OF_SMALL_MOLECULES	0.06514096	566	1.566e-07	3.382e-04	SLC1A1.12 AQP11.35 MTPP.36 ALB.51 SLC22A1.62 SLC5A8.64
WP_NRF2_PATHWAY	0.16050922	92	1.077e-07	3.382e-04	TXNRD3.58 SLC5A8.64 SLC5A12.67 MGST3.80 SLC2A9.173 SLC6A15.194
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS	0.14702944	95	7.612e-07	1.232e-03	SLC1A1.12 SLC5A8.64 SLC5A12.67 SLC6A12.119 SLC36A2.138 SLC4A9.176
WP_NUCLEAR_RECEPTORS_METAPATHWAY	0.09475113	222	1.257e-06	1.628e-03	EHHADH.29 ABCB11.56 TXNRD3.58 SLC5A8.64 SLC5A12.67 MGST3.80
REACTOME_TRANSPORT_OF_BILE_SALTS_AND_ORG	0.15728318	77	1.880e-06	1.951e-03	SLC22A1.62 SLC6A12.119 SLC6A15.194 SLC13A2.205 SLC6A18.429 SLC6A6.478
REACTOME_SLC_TRANSPORTER_DISORDERS	0.14901738	85	2.109e-06	1.951e-03	SLC1A1.12 SLC36A2.138 NUP133.147 SLC2A9.173 SLC26A4.296 SLC12A6.305
CAIRO_HEPATOBLASTOMA_DN	0.09447100	196	5.490e-06	4.444e-03	SLC1A1.12 GYS2.34 ABCB11.56 SLC22A1.62 GCDH.78 HP.90
WP_TRANSULFURATION_ONECARBON_METABOLISM	0.18126636	51	7.645e-06	5.501e-03	BCAT2.28 CHDH.127 GNM1.374 PHGDH.301 GPX4.371 TYMS.468
FEVR_CTNNB1_TARGETS_UP	0.05261648	502	6.327e-05	3.854e-02	CLP5.1 SLC1A1.12 ENTDP5.20 SPTA1.33 AQP11.35 MTPP.36
LEE_LIVER_CANCER_DENA_DN	0.14481559	63	7.142e-05	3.854e-02	EHHADH.29 PNLP.37 GCDH.78 IL1RAP.97 CYP7A1.131 ABCD3.167
HSIAO_LIVER_SPECIFIC_GENES	0.08571511	183	6.695e-05	3.854e-02	GYS2.34 MTPP.36 ALB.51 LGAL.54.60 TM4SF4.86 HP.90
REACTOME_BIOLOGICAL_OXIDATIONS	0.10142111	123	1.058e-04	4.952e-02	CYP11A1.4 PODXL2.66 MGST3.80 CYP7A1.131 COMT.180 CYP2F1.253
REACTOME_METABOLISM_OF_LIPIDS	0.04752273	581	1.071e-04	4.952e-02	CYP11A1.4 FAR2.188 PI3K.25 EHHAADH.29 MORC2.30 ALB.51
REACTOME_SIGNALING_BY_WNT	-0.08640085	161	1.614e-04	6.967e-02	WNT4.25 DVL2.140 RAC2.161 H3C9.239 VPS26A.283 FRAT1.358
REACTOME_AMINO_ACID_TRANSPORT_ACROSS_THE	0.19410420	30	2.348e-04	9.504e-02	SLC6A12.119 SLC36A2.138 SLC6A15.194 SLC6A18.429 SLC6A6.478 SLC43A1.553
WP_Q13_COPY_NUMBER_VARIATION_SYNDROME	0.16072059	43	2.680e-04	1.021e-01	MERTK.70 IL1RL2.87 IL1RAP.97 ZC3H6.245 TTL.373 PSD4.380
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.20544983	26	2.889e-04	1.040e-01	CYP11A1.4 CYP7A1.131 COMT.180 HSD17B3.574 HSD17B8.679 SULT2B1.692
WP_OXIDATIVE_STRESS_RESPONSE	0.20027966	27	3.170e-04	1.081e-01	NOX1.213 TXNRD1.282 NQO1.374 MGST1.402 CYP1A1.695 HMOX1.923
WP_METAPATHWAY_BIOTRANSFORMATION_PHASE_I	0.1068406	92	4.049e-04	1.311e-01	CYP11A1.4 MGST3.80 CYP7A1.131 COMT.180 CYP2F1.253 GPX4.371
HOSHIDA_LIVER_CANCER_SUBCLASS_S3	0.06908213	210	5.853e-04	1.805e-01	SLC23A1.11 EHHAADH.29 SLC6A12.119 MMUT.175 AGXT.219 VD.264
REACTOME_DIGESTION_OF_DIETARY_LIPID	0.48407682	4	7.991e-04	2.070e-01	CLP5.1 PNLP.37 PNLP.37 PNLP.37 PNLP.37 PNLP.37 PNLP.37 PNLP.37 PNLP.37
BLAIKOC_ALZHEIMERS_DISEASE_DN	-0.03617836	765	7.857e-04	2.070e-01	PTEN.5 ATP6V1C1.10 STXBP1.64 RALB.67 PCDH11X.69 NDUFAR.88
CHIANG_LIVER_CANCER_SUBCLASS_PROLIFERATI	0.08421263	136	7.164e-04	2.070e-01	EHHADH.29 GYS2.34 MTPP.36 ABCB11.56 SLC22A1.62 GCDH.78
WOO_LIVER_CANCER_RECURRENCE_DN	0.12599558	60	7.449e-04	2.070e-01	MTPP.36 GCDH.78 SLC2A9.173 AAS.238 ACADS.561 SARDB.634
REACTOME_DISORDERS_OF_TRANSMEMBRANE_TRAN	0.08259665	133	1.031e-03	2.568e-01	SLC1A1.12 ABCB11.56 SLC36A2.138 NUP133.147 SLC2A9.173 SLC26A4.296
STARK_HYPOCAEMUS_22Q11_DELETION_DN	0.22904477	17	1.079e-03	2.588e-01	HIRA.61 TRMT2A.169 COMT.180 ARVCF.608 MRPL.407 ZDHHC.933
REACTOME_HEME_DEGRADATION	0.29579473	10	1.200e-03	2.658e-01	ALB.51 FABP1.1179 HMOX1.923 UGT1A1.925 SLC01B1.1202 BLVRB.1531
REACTOME_BETA_CATENIN_INDEPENDENT_WNT_SI	-0.10463927	80	1.231e-03	2.658e-01	WNT4.25 DVL2.140 RAC2.161 FZD6.407 AGO3.414 RAC1.451
REACTOME_SENSORY_PERCEPTION_OF_TASTE	0.17407477	29	1.181e-03	2.658e-01	TAS2R41.71 TAS2R4.208 SCNN1G.358 TAS1R1.385 SCNN1B.406 TAS1R3.521
WP_FOCAL_ADHESION	-0.07403951	158	1.359e-03	2.825e-01	PTEN.5 BIRC3.58 CRK.158 RAC2.161 MAPK8.211 VEGFD.282
BERTUCCI_MEDULLARY_VS_DUCTAL_BREAST_CANC	-0.08427575	121	1.396e-03	2.825e-01	GUCY1A2.127 UHRF1.131 SNRP.218 ADAMTS12.267 SPOCK1.286 ACTN1.4699.5
ANDERSEN_LIVER_CANCER_KRT19_DN	0.12320233	54	1.678e-03	2.841e-01	GNMT.174 PCYOX1.287 SELENBP1.414 PTGR1.540 HPD.760 AOX1.779
CARRILLORREIXACH_HEPATOBLASTOMA_VS_NORMAL	0.03351551	785	1.642e-03	2.841e-01	CYP11A1.4 HMCE5.6 SLC1A1.12 EHHAADH.29 GYS2.34 ABCB11.56
KEGG_TASTE_TRANSDUCTION	0.17611416	27	1.543e-03	2.841e-01	TAS2R41.71 TAS2R4.208 SCNN1G.358 TAS1R1.385 SCNN1B.406 ADCY4.460
REACTOME_TRANSLATION	-0.06703756	189	1.535e-03	2.841e-01	RPS3C.32 MRPL39.78 SEC61B.106 MRPL37.139 FARSA.194 DAP2.313
REACTOME_DIGESTION_AND_ABSORPTION	0.23391998	15	1.711e-03	2.841e-01	CLP5.1 PNLP.37 PNLP.37 PNLP.37 PNLP.37 PNLP.37 PNLP.37 PNLP.37 PNLP.37
YOSHIMURA_MAPK8_TARGETS_UP	0.03201275	877	1.546e-03	2.841e-01	CLP5.1 SLC1A1.12 GYS2.34 PNLP.37 RBP2.57 LGAL.54.60
KEGG_FOCAL_ADHESION	-0.07267361	158	1.666e-03	2.841e-01	PTEN.5 BIRC3.58 CRK.158 RAC2.161 MAPK8.211 VEGFD.282
VERNOCHET_ADIPOGENESIS	0.24823034	13	1.944e-03	2.861e-01	HP.90 APCDD1.157 MCR2.224 CYP2F1.253 FFA2.426 RARRES2.1197

DisGeNET Top pathways by non-permutation

Kidney Calculi	0.17864677	51	1.042e-05	1.018e-01	PTH.77 CLDN16.103 SLC2A9.173 SLC13A2.205 AGXT.219 SLC4A1.312
Cystic Fibrosis	0.06158866	374	5.133e-05	1.671e-01	BP1FB1.2 SGSM3.46 ALB.51 NPCL1.116 MYO8B.125 NR12.133
Hyperkalemia	0.30300147	15	4.865e-05	1.671e-01	CYP11A1.4 SCNN1G.358 SCNN1B.406 SCNN1A.653 CA12.836 WNK1.975
Hypertensive disease	0.03945789	918	7.795e-05	1.903e-01	CYP11A1.4 ALB.51 TXNRD3.58 PTH.77 HP.90 PRKCD.91
Anemia, hereditary spherocytic hemolytic	0.39463988	5	2.243e-03	7.301e-01	SPTA1.33 SLC4A1.312 ANK1.604 EPB4.6247 D2860 SPTB.3398 NA
Hyperlipidemia, Familial Combined	0.11397288	62	1.941e-03	7.301e-01	MTPP.36 ALB.51 CYP7A1.131 FADS3.269 TXNIP.363 CUX1.610
Hypoxia-Ischemia, Brain	0.31688639	8	1.913e-03	7.301e-01	NOS2.197 TNF.1163 HP1A.1560 IRAX1.1766 IL6.1791 VEGFA.1801
PSEUDOHYPOLADOSTERONISM, TYPE IID	0.35584085	6	2.541e-03	7.301e-01	SLC12A3.356 WNK1.975 PRPF4B.1079 TRPV5.1660 WNK4.3177 NEK1.3567
22q11 partial monosomy syndrome	0.28010784	10	2.164e-03	7.301e-01	HIRA.61 COMT.180 ARVCF.608 UFD1.1138 GPIIB.2250 PI4KA.2393
Abnormality of aortic arch	0.27826471	10	2.314e-03	7.301e-01	HIRA.61 COMT.180 ARVCF.608 UFD1.1138 GPIIB.2250 MCTP2.2435
Benign Prostatic Hyperplasia	0.05466704	266	2.304e-03	7.301e-01	CYP11A1.4 GPRC5A.24 PROM1.40 TXNRD3.58 PGC.65 HP.90
Benign recurrent intrahepatic cholestasis	0.37988371	6	1.272e-03	7.301e-01	ABCB11.56 GTCLC3.793 GGT2.1616 ATP8B1.1886 GGT1.2045 GTCLC1.2577
Cardiovascular Diseases	0.03874618	561	1.999e-03	7.301e-01	MTPP.36 ALB.51 TXNRD3.58 PTH.77 HP.90 NPCL1.116
Cholelsthiasis	0.06710457	229	5.054e-04	7.301e-01	SLC23A1.11 ABCB11.56 SLC22A1.62 HP.90 NPCL1.116 CYP7A1.131
Chronic Kidney Diseases	0.05968537	263	1.138e-03	7.301e-01	ALB.51 SLC22A1.62 PTH.77 HP.90 CLDN16.103 CHDH.127
Classical galactosemia	-0.25887603	14	7.995e-04	7.301e-01	GAL.16 S100A1.21 GALK1.228 LGALS1.259 DMD.324 TGFBI.4699.5
Fatty Liver	0.05037213	312	2.409e-03	7.301e-01	SUGP1.27 MTPP.36 SGSM3.46 ALB.51 HP.90 NPCL1.116
Hereditary spherocytosis	0.20080870	21	9.145e-04	7.301e-01	SPTA1.33 SLC4A1.312 ANK1.604 CAD.730 SLC26A8.768 G6PD.828
Hyperemia	0.19775945	25	6.241e-04	7.301e-01	PTH.77 NOS2.197 ADRB1.559 TPGS2.587 HRH1.606 SERPINE1.739
Hypocalcemia	0.13895742	40	2.228e-03	7.301e-01	HIRA.61 PTH.77 CLDN16.103 CHDH.127 COMT.180 SLC4A1.312
Impaired T cell function	0.23614517	15	1.547e-03	7.301e-01	HIRA.61 COMT.180 ARVCF.608 UFD1.1138 TNFRSF13B.1510 SLC39A4.1680
Inborn Errors of Metabolism	0.11224800	61	2.464e-03	7.301e-01	GDH.78 HEXB.164 AGXT.219 TXNRD1.282 GLA.314 SELENBP1.414
Intestinal Polyps	0.27198546	27	1.377e-03	7.301e-01	CNNM4.204 PLA2G1B.225 NAGLU.349 GUCYA2A.558 TPGS2.587 NR4A2.622
Left-Sided Heart Failure	0.09029723	94	2.542e-03	7.301e-01	ALB.51 PTH.77 NOS2.197 NOX1.213 ATP2A1.216 GHRH.252
Liver neoplasms	0.03089985	870	2.531e-03	7.301e-01	SUGP1.27 MTPP.36 PROM1.40 TIGAR.50 ALB.51 ABCB11.56
Nephrolithiasis	0.11583076	76	4.925e-04	7.301e-01	PTH.77 CLDN16.103 CDG73.156 SLC2A9.173 SLC13A2.205 AGXT.219
Persistent truncus arteriosus	0.20222111	19	1.273e-03	7.301e-01	HIRA.61 GJA5.145 COMT.180 ARVCF.608 DLL4.731 MAPK1.882
Recurrent infections	0.13487133	48	1.242e-03	7.301e-01	PRKCD.91 MANBA.104 MC2R.224 SLC46A1.524 CARD11.651 VAMP1.664
Reticulocytosis	0.21942946	20	6.911e-04	7.301e-01	SEC23B.31 SLC4A1.312 KCNN4.547 ANK1.604 G6PD.828 RHAG.855
Sea-Blue Histiocyte Syndrome	0.41437531	5	1.333e-03	7.301e-01	NPCL1.116 SMPD1.276 NPC2.467 GLB1.1134 APOE.3350 NA
Seborrheic dermatitis	0.22741693	18	8.401e-04	7.301e-01	HIRA.61 COMT.180 ZNF750.518 ARVCF.608 UFD1.1138 TNF.1163
Spherocytosis	0.39463988	5	2.243e-03	7.301e-01	SPTA1.33 SLC4A1.312 ANK1.604 EPB4.6247 SPTB.3398 NA
Taste wetness	0.25991207	12	1.872e-03	7.301e-01	TAS1R1.385 TAS1R2.321 SLC2A2.899 LPAR2.1039 TAS1R2.1400 SLC2A2.1415
Tetany	0.21303369	19	1.311e-03	7.301e-01	HIRA.61 PTH.77 CLDN16.103 COMT.180 SLC12A3.356 KCNJ1.498
Adenocarcinoma of rectum	0.17318705	24	3.390e-03	7.560e-01	MYD88.125 CDC159 ENTDP1.166 SLC12A9.271 TYMS.468 TP53BP1.470
Cholangiolocellular Carcinoma	0.41798942	4	3.789e-03	7.560e-01	PROM1.40 ALB.51 CRKL1.1518 MTHFR.2476 NA NA
Drug-Induced Liver Disease	0.06206984	183	3.947e-03	7.560e-01	IL1R2.17 ALB.51 ABCB11.56 LRAF.1122 NR12.133 ATR.135
Epididymolysis Bullosa	-0.15632613	30	3.061e-03	7.560e-01	KLHL24.83 KRT14.150 ITGB4.4699.5 TP53.4699.5 LAMC2.4699.5 D2.4699.5
Essential Hypertension	0.05963516	199	3.908e-03	7.560e-01	ALB.51 PTH.77 HP.90 GJA5.145 NOS2.197 NOX1.213
Hyperactive renin-angiotensin system	0.29451018	8	3.924e-03	7.560e-01	SCNN1G.358 SCNN1B.406 KCNJ1.349 SCNN1A.653 NR3C2.1184 SLC26A3.1580